



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Coordinate Systems-Examples

Example 1

Given point $P(-2, 6, 3)$ and vector $\mathbf{A} = y\mathbf{a}_x + (x + z)\mathbf{a}_y$, express P and $\mathbf{A}$ in cylindrical and spherical coordinates. Evaluate $\mathbf{A}$ at P in the Cartesian, cylindrical, and spherical systems.

Solution

At point P : $x = -2, y = 6, z = 3$. Hence,

$$\rho = \sqrt{x^2 + y^2} = \sqrt{4 + 36} = 6.32$$

$$\phi = \tan^{-1} \frac{y}{x} = \tan^{-1} \frac{6}{-2} = 108.43^\circ$$

$$z = 3$$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{4 + 36 + 9} = 7$$

$$\theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z} = \tan^{-1} \frac{\sqrt{40}}{3} = 64.62^\circ$$

Thus,

$$P(-2, 6, 3) = P(6.32, 108.43^\circ, 3) = P(7, 64.62^\circ, 108.43^\circ)$$

In the Cartesian system, $\mathbf{A}$ at P is

$$\mathbf{A} = 6\mathbf{a}_x + \mathbf{a}_y$$

For vector $\mathbf{A}$, $A_x = y, A_y = x + z, A_z = 0$. Hence, in the cylindrical system

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y \\ x + z \\ 0 \end{bmatrix}$$

or

$$\begin{aligned}
A_\rho &= y \cos \phi + (x + z) \sin \phi \\
A_\phi &= -y \sin \phi + (x + z) \cos \phi \\
A_z &= 0
\end{aligned}$$

But $x = \rho \cos \phi$, $y = \rho \sin \phi$, and substituting these yields

$$\begin{aligned}
\mathbf{A} = (A_\rho, A_\phi, A_z) &= [\rho \cos \phi \sin \phi + (\rho \cos \phi + z) \sin \phi] \mathbf{a}_\rho \\
&\quad + [-\rho \sin^2 \phi + (\rho \cos \phi + z) \cos \phi] \mathbf{a}_\phi
\end{aligned}$$

At P

$$\rho = \sqrt{40}, \quad \tan \phi = \frac{6}{-2}$$

Hence,

$$\begin{aligned}
\cos \phi &= \frac{-2}{\sqrt{40}}, \quad \sin \phi = \frac{6}{\sqrt{40}} \\
\mathbf{A} &= \left[\sqrt{40} \cdot \frac{-2}{\sqrt{40}} \cdot \frac{6}{\sqrt{40}} + \left(\sqrt{40} \cdot \frac{-2}{\sqrt{40}} + 3 \right) \cdot \frac{6}{\sqrt{40}} \right] \mathbf{a}_\rho \\
&\quad + \left[-\sqrt{40} \cdot \frac{36}{40} + \left(\sqrt{40} \cdot \frac{-2}{\sqrt{40}} + 3 \right) \cdot \frac{-2}{\sqrt{40}} \right] \mathbf{a}_\phi \\
&= \frac{-6}{\sqrt{40}} \mathbf{a}_\rho - \frac{38}{\sqrt{40}} \mathbf{a}_\phi = -0.9487 \mathbf{a}_\rho - 6.008 \mathbf{a}_\phi
\end{aligned}$$

Similarly, in the spherical system

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} y \\ x + z \\ 0 \end{bmatrix}$$

or

$$\begin{aligned}
A_r &= y \sin \theta \cos \phi + (x + z) \sin \theta \sin \phi \\
A_\theta &= y \cos \theta \cos \phi + (x + z) \cos \theta \sin \phi \\
A_\phi &= -y \sin \phi + (x + z) \cos \phi
\end{aligned}$$

But $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, and $z = r \cos \theta$. Substituting these yields

$$\begin{aligned}
\mathbf{A} &= (A_r, A_\theta, A_\phi) \\
&= r[\sin^2 \theta \cos \phi \sin \phi + (\sin \theta \cos \phi + \cos \theta) \sin \theta \sin \phi] \mathbf{a}_r \\
&\quad + r[\sin \theta \cos \theta \sin \phi \cos \phi + (\sin \theta \cos \phi + \cos \theta) \cos \theta \sin \phi] \mathbf{a}_\theta \\
&\quad + r[-\sin \theta \sin^2 \phi + (\sin \theta \cos \phi + \cos \theta) \cos \phi] \mathbf{a}_\phi
\end{aligned}$$

At P

$$r = 7, \quad \tan \phi = \frac{6}{-2}, \quad \tan \theta = \frac{\sqrt{40}}{3}$$

Hence,

$$\begin{aligned}
 \cos \phi &= \frac{-2}{\sqrt{40}}, \quad \sin \phi = \frac{6}{\sqrt{40}}, \quad \cos \theta = \frac{3}{7}, \quad \sin \theta = \frac{\sqrt{40}}{7} \\
 \mathbf{A} &= 7 \cdot \left[\frac{40}{49} \cdot \frac{-2}{\sqrt{40}} \cdot \frac{6}{\sqrt{40}} + \left(\frac{\sqrt{40}}{7} \cdot \frac{-2}{\sqrt{40}} + \frac{3}{7} \right) \cdot \frac{\sqrt{40}}{7} \cdot \frac{6}{\sqrt{40}} \right] \mathbf{a}_r \\
 &\quad + 7 \cdot \left[\frac{\sqrt{40}}{7} \cdot \frac{3}{7} \cdot \frac{6}{\sqrt{40}} \cdot \frac{-2}{\sqrt{40}} + \left(\frac{\sqrt{40}}{7} \cdot \frac{-2}{\sqrt{40}} + \frac{3}{7} \right) \cdot \frac{3}{7} \cdot \frac{6}{\sqrt{40}} \right] \mathbf{a}_\theta \\
 &\quad + 7 \cdot \left[\frac{-\sqrt{40}}{7} \cdot \frac{36}{40} + \left(\frac{\sqrt{40}}{7} \cdot \frac{-2}{\sqrt{40}} + \frac{3}{7} \right) \cdot \frac{-2}{\sqrt{40}} \right] \mathbf{a}_\phi \\
 &= \frac{-6}{7} \mathbf{a}_r - \frac{18}{7\sqrt{40}} \mathbf{a}_\theta - \frac{38}{\sqrt{40}} \mathbf{a}_\phi \\
 &= -0.8571 \mathbf{a}_r - 0.4066 \mathbf{a}_\theta - 6.008 \mathbf{a}_\phi
 \end{aligned}$$

Note that $|\mathbf{A}|$ is the same in the three systems; that is,

$$|\mathbf{A}(x, y, z)| = |\mathbf{A}(\rho, \phi, z)| = |\mathbf{A}(r, \theta, \phi)| = 6.083$$

Example 2

Two uniform vector fields are given by $\mathbf{E} = -5\mathbf{a}_\rho + 10\mathbf{a}_\phi + 3\mathbf{a}_z$ and $\mathbf{F} = \mathbf{a}_\rho + 2\mathbf{a}_\phi - 6\mathbf{a}_z$. Calculate

- $|\mathbf{E} \times \mathbf{F}|$
- The vector component of $\mathbf{E}$ at $P(5, \pi/2, 3)$ parallel to the line $x = 2, z = 3$
- The angle that $\mathbf{E}$ makes with the surface $z = 3$ at P

Solution

$$\begin{aligned}
 \text{(a) } \mathbf{E} \times \mathbf{F} &= \begin{vmatrix} \mathbf{a}_\rho & \mathbf{a}_\phi & \mathbf{a}_z \\ -5 & 10 & 3 \\ 1 & 2 & -6 \end{vmatrix} \\
 &= (-60 - 6)\mathbf{a}_\rho + (3 - 30)\mathbf{a}_\phi + (-10 - 10)\mathbf{a}_z \\
 &= (-66, -27, -20) \\
 |\mathbf{E} \times \mathbf{F}| &= \sqrt{66^2 + 27^2 + 20^2} = 74.06
 \end{aligned}$$

- (b) Line $x = 2, z = 3$ is parallel to the y -axis, so the component of $\mathbf{E}$ parallel to the given line is

$$(\mathbf{E} \cdot \mathbf{a}_y) \mathbf{a}_y$$

But at $P(5, \pi/2, 3)$

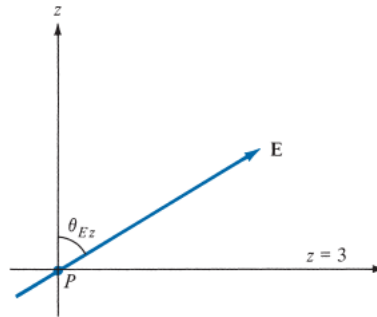
$$\begin{aligned}
 \mathbf{a}_y &= \sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi \\
 &= \sin \pi/2 \mathbf{a}_\rho + \cos \pi/2 \mathbf{a}_\phi = \mathbf{a}_\rho
 \end{aligned}$$

Therefore,

$$(\mathbf{E} \cdot \mathbf{a}_y) \mathbf{a}_y = (\mathbf{E} \cdot \mathbf{a}_\rho) \mathbf{a}_\rho = -5\mathbf{a}_\rho \quad (\text{or } -5\mathbf{a}_y)$$

- (c) Since the z -axis is normal to the surface $z = 3$, we can use the dot product to find the angle between the z -axis and $\mathbf{E}$, as shown

$$\begin{aligned}
 \mathbf{E} \cdot \mathbf{a}_z &= |\mathbf{E}|(1) \cos \theta_{Ez} \rightarrow 3 = \sqrt{134} \cos \theta_{Ez} \\
 \cos \theta_{Ez} &= \frac{3}{\sqrt{134}} = 0.2592 \rightarrow \theta_{Ez} = 74.98^\circ
 \end{aligned}$$



Hence, the angle between $z = 3$ and $\mathbf{E}$ is

$$90^\circ - \theta_{Ez} = 15.02^\circ$$